SP-14 - Green - Chess AI

CS 4850 – Section 03 – 11/4/2025

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Team Members:

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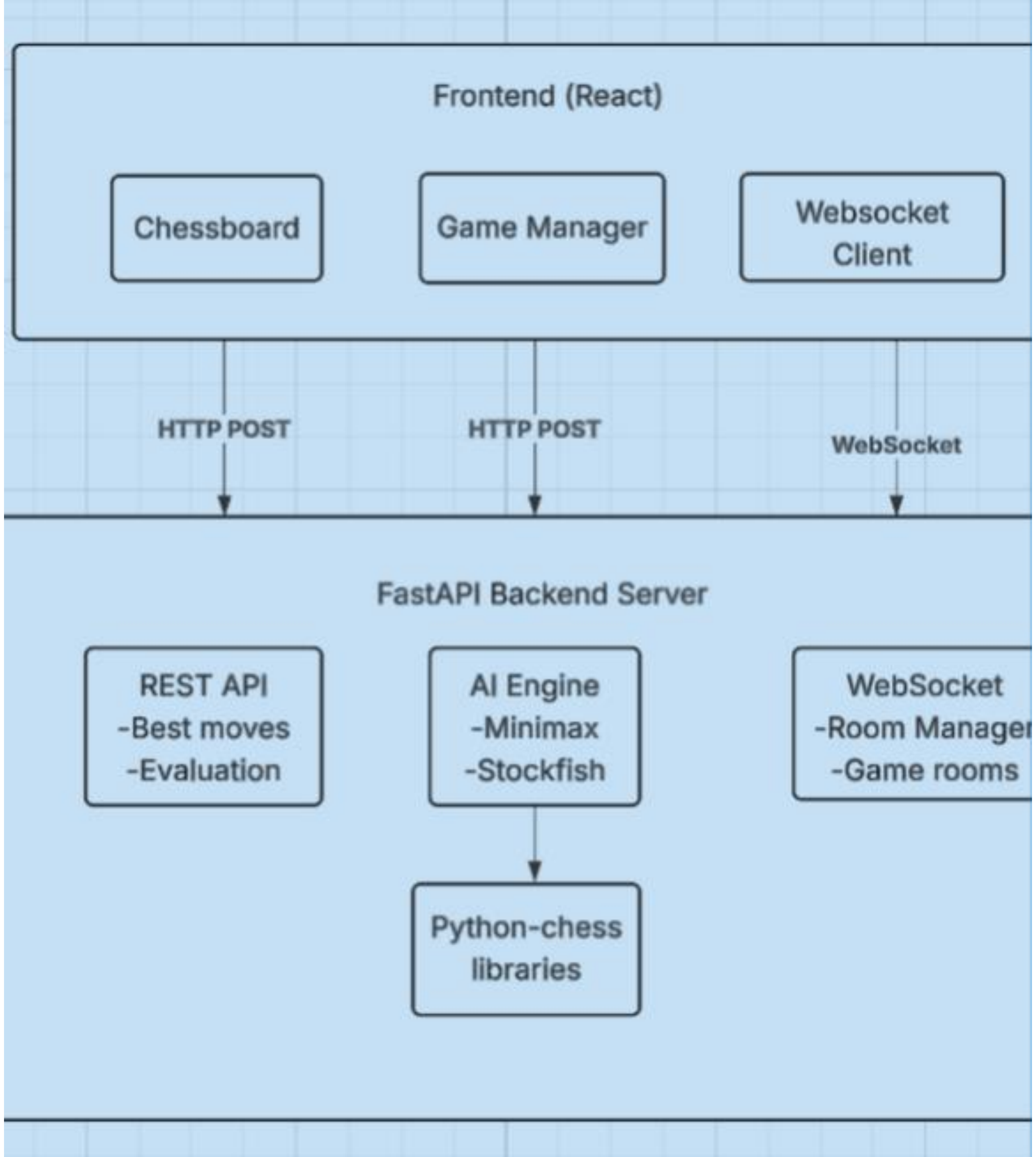
Problem statements

- To create a chess game with robust features with a quality of provided libraries.

- Create a self-playing AI Chess Engine that offers both player vs player and player vs AI game modes.

- Increase functionality
  - Multiplayer(locally or online)
  - Self-play AI vs AI
  - Web development
  - Piece customization

Chess Game with AI Project

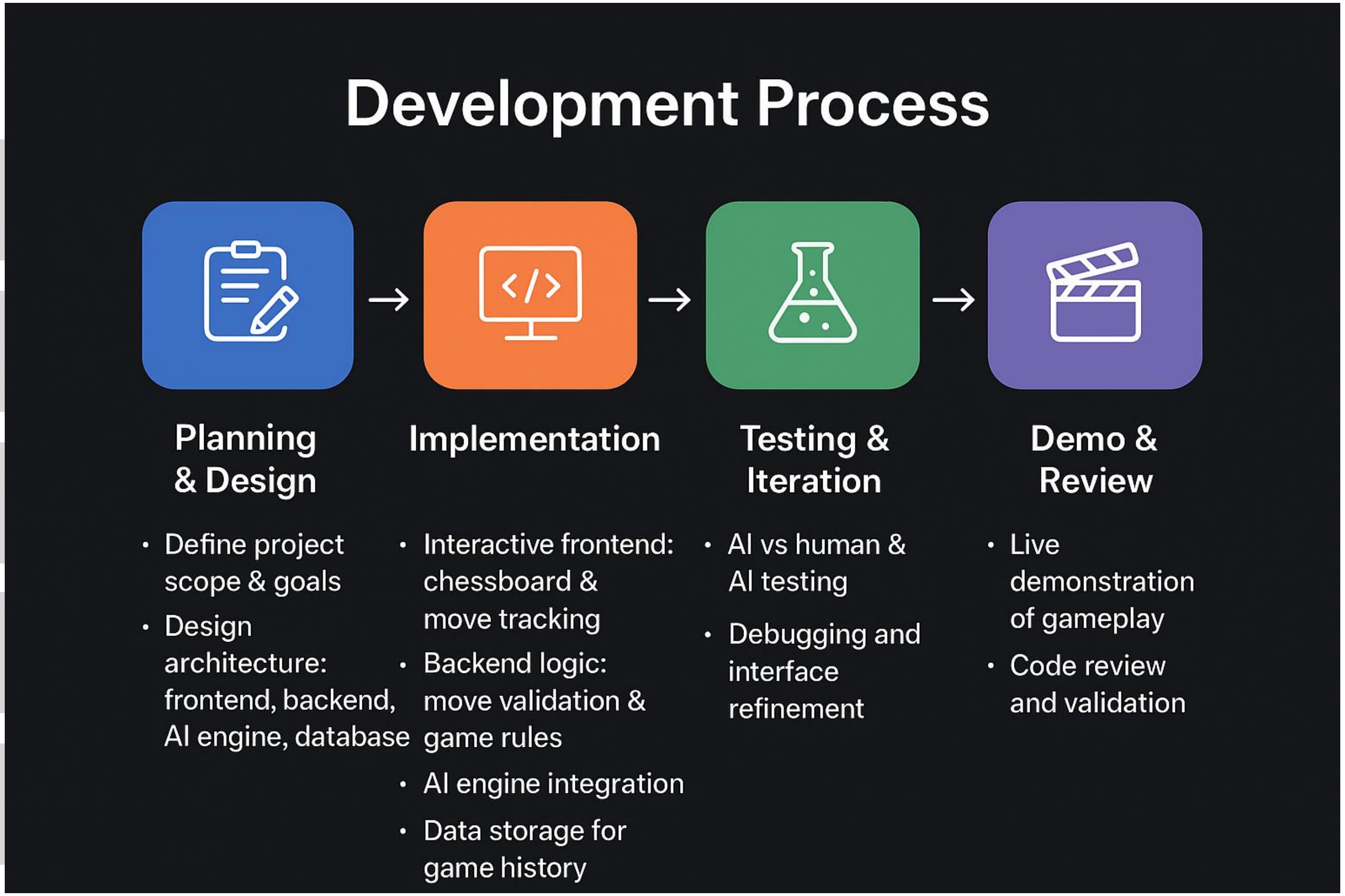


Tech Stack

- Web App
- React Framework js Frontend
  - React Chessboard- visualization
  - Chess.js- Handle chess game logic
  - Vite- Fast development server
- Python3 Backend
  - FastAPI- web framework for HTTPS request
  - Python Chess- handle chess game logic and AI engine move generation
  - Uvicorn- ASGI server to run FastAPI

Process Methodology: Agile CI/CD

- Planning 📝
- Design 🔬
- Implementation 💻
- Testing 🕷️
- Demo 🎬



Project Setup ⚙️

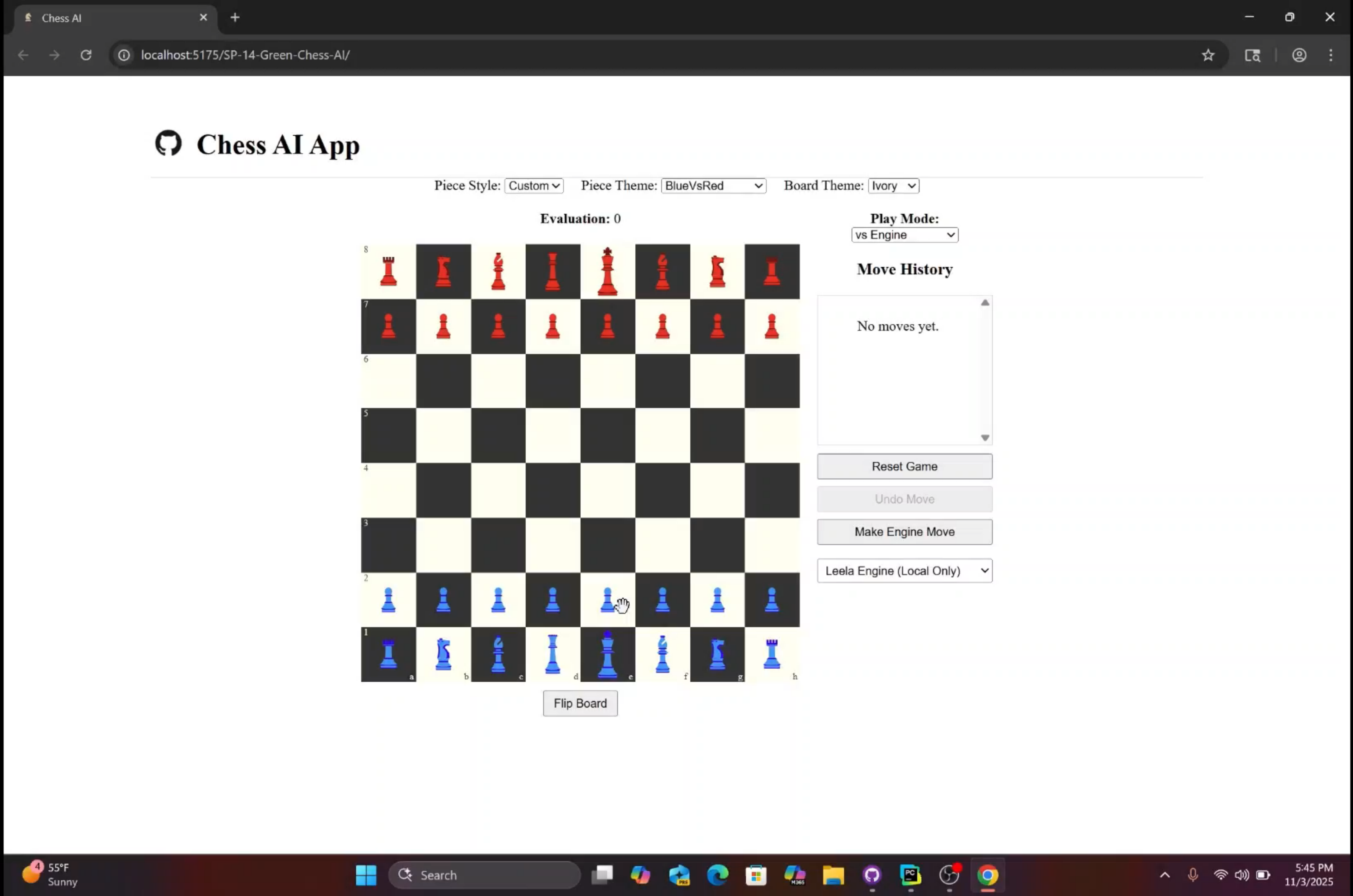
Prerequisites:

- Python 3.8+ & Node.js 14+
- npm & Git installed
- IDE: VS Code or PyCharm
- Web browser to access app

Start player vs AI, AI vs AI demo

AI Engines

- **Custom Minimax**- A classic game theory algorithm with depth of 2 to 3 search. Can be used to find best move 2-3 complete turns in the future.
- **Leela Chess Zero (Lc0) -** uses a deep neural network trained on millions of games played. Leela recognizes positional and strategic patterns by running 800 Monte Carlo simulations per move and finds the move with the highest win probability.
- **Stockfish**- uses alpha-beta pruning to search 8-15 complete turns in the future.


Chess AI
localhost:5175/SP-14-Green-Chess-AI/
Chess AI App
Piece Style: Custom
Piece Theme: BlueVsRed
Board Theme: Ivory
Evaluation: 0
Play Mode:
vs Engine
Move History
No moves yet.
Reset Game
Undo Move
Make Engine Move
Leela Engine (Local Only)
Flip Board
55°F
Sunny
Search
5:45 PM
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Basic Game Modes

Local – 2 Player

Source: Saint Louis Chess Club. “Clutch Chess Champions Showdown.” Event photography. ClutchChess.com

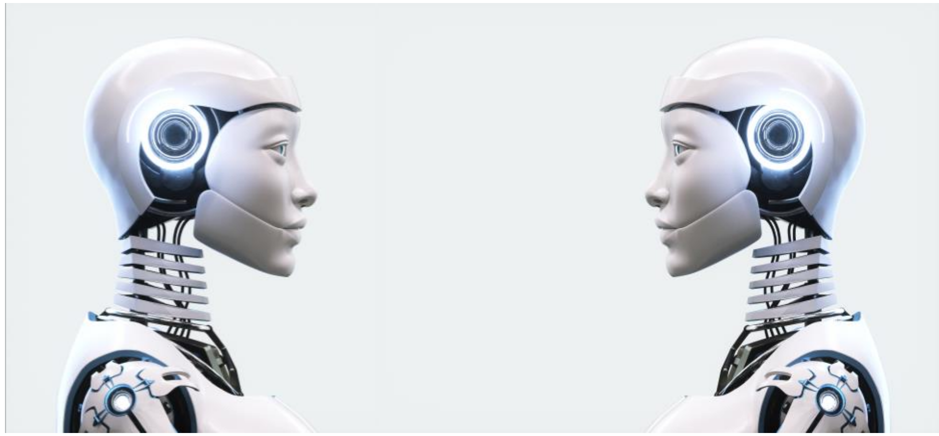
Player vs AI

Advanced Game Modes

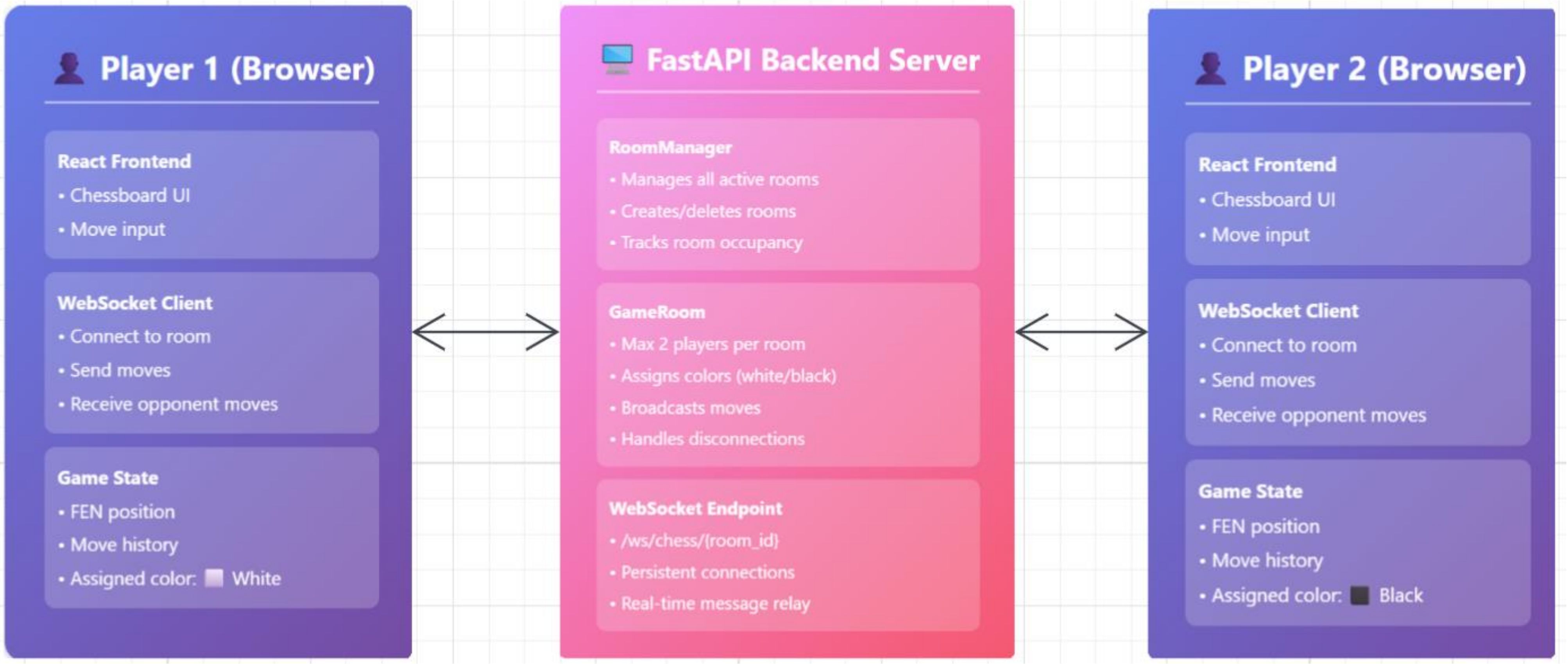
Online Multiplayer

AI vs AI

Source: EWC chess tournamnet. *"Magnus Carlsen wins first ever edition of Chess Esports World Cup " Event photography. esportsworldcup.com*

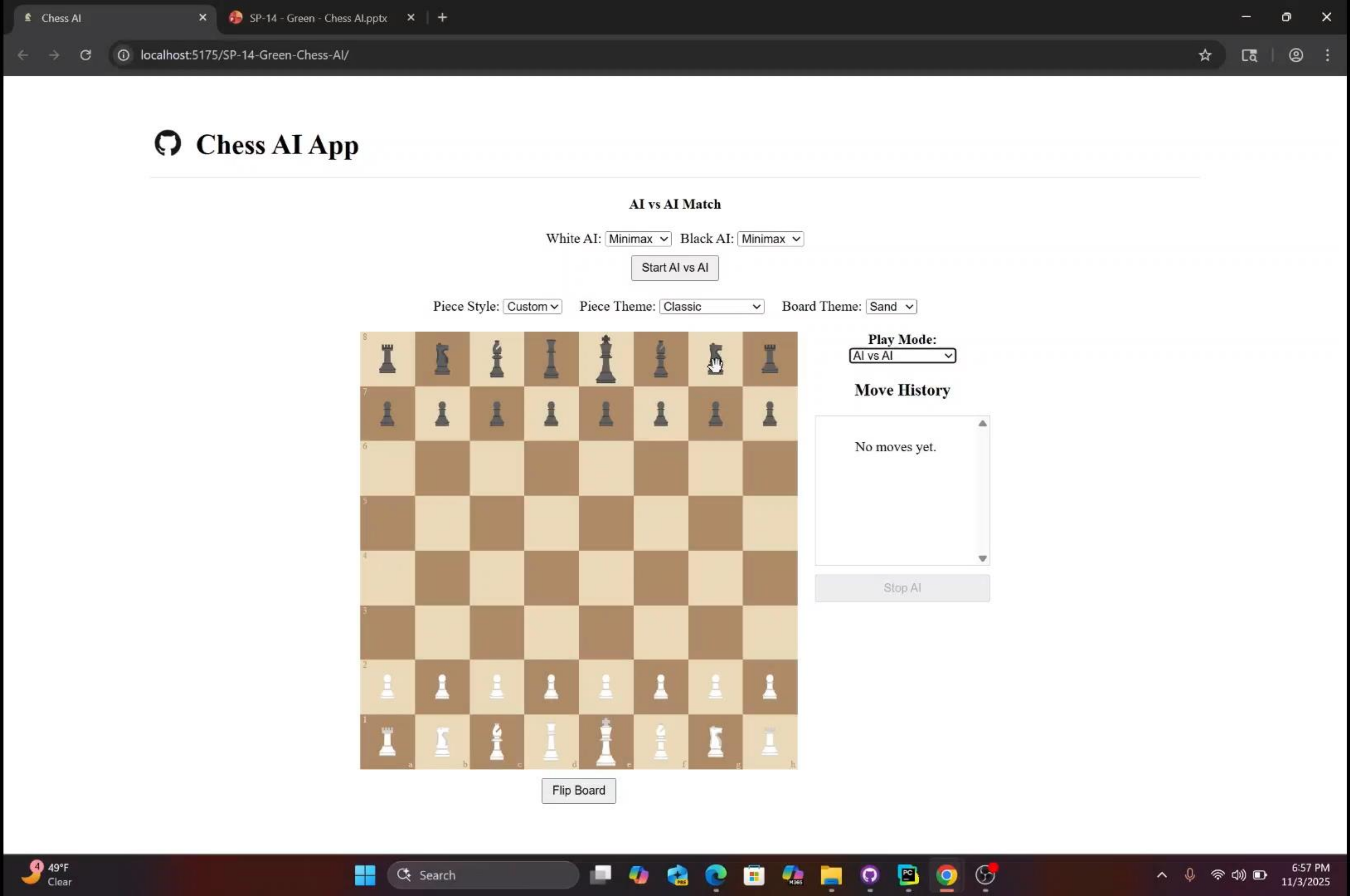


Online Multiplayer Architecture

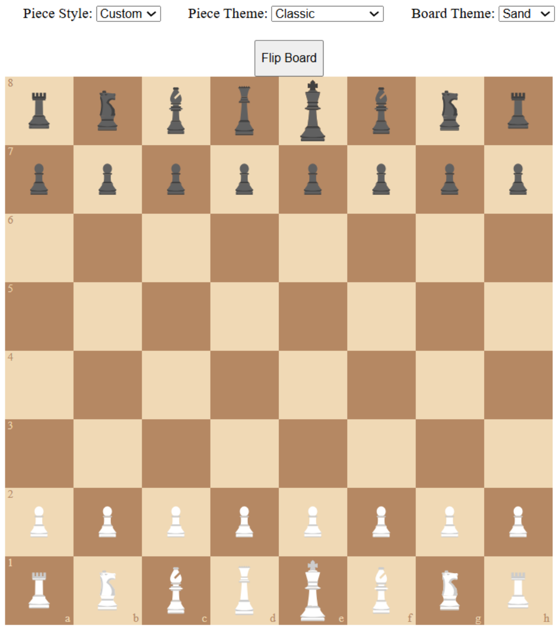


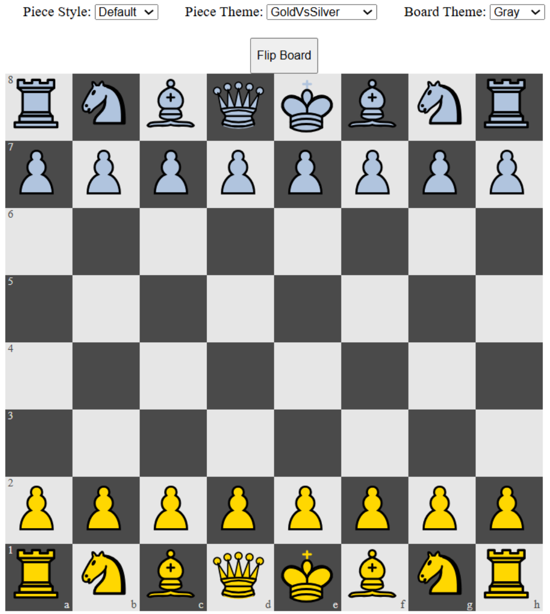
Why?

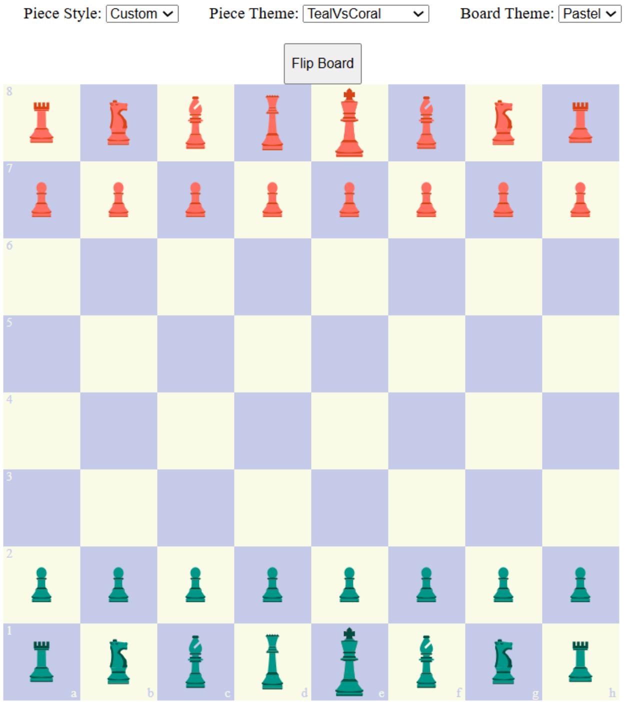
- WebSocket:
    - Real-time bi-directional communication
    - Instant move synchronization
    - Low latency
    - Able to create a game room architecture


Chess AI
SP-14 - Green - Chess AI.pptx
localhost:5175/SP-14-Green-Chess-AI/
Chess AI App
AI vs AI Match
White AI: Minimax
Black AI: Minimax
Start AI vs AI
Piece Style: Custom
Piece Theme: Classic
Board Theme: Sand
Play Mode:
AI vs AI
Move History
No moves yet.
Stop AI
Flip Board
49°F
Clear
Search
6:57 PM
11/3/2025

Themes and Style Customization







Experimental AI: (Monte-Carlo)

- Core Concept
  - Algorithm for making optimal decisions in games with massive decision spaces
  - Builds a search tree incrementally by running random simulations
  - Balances exploration (trying new moves) vs exploitation (using known good moves)

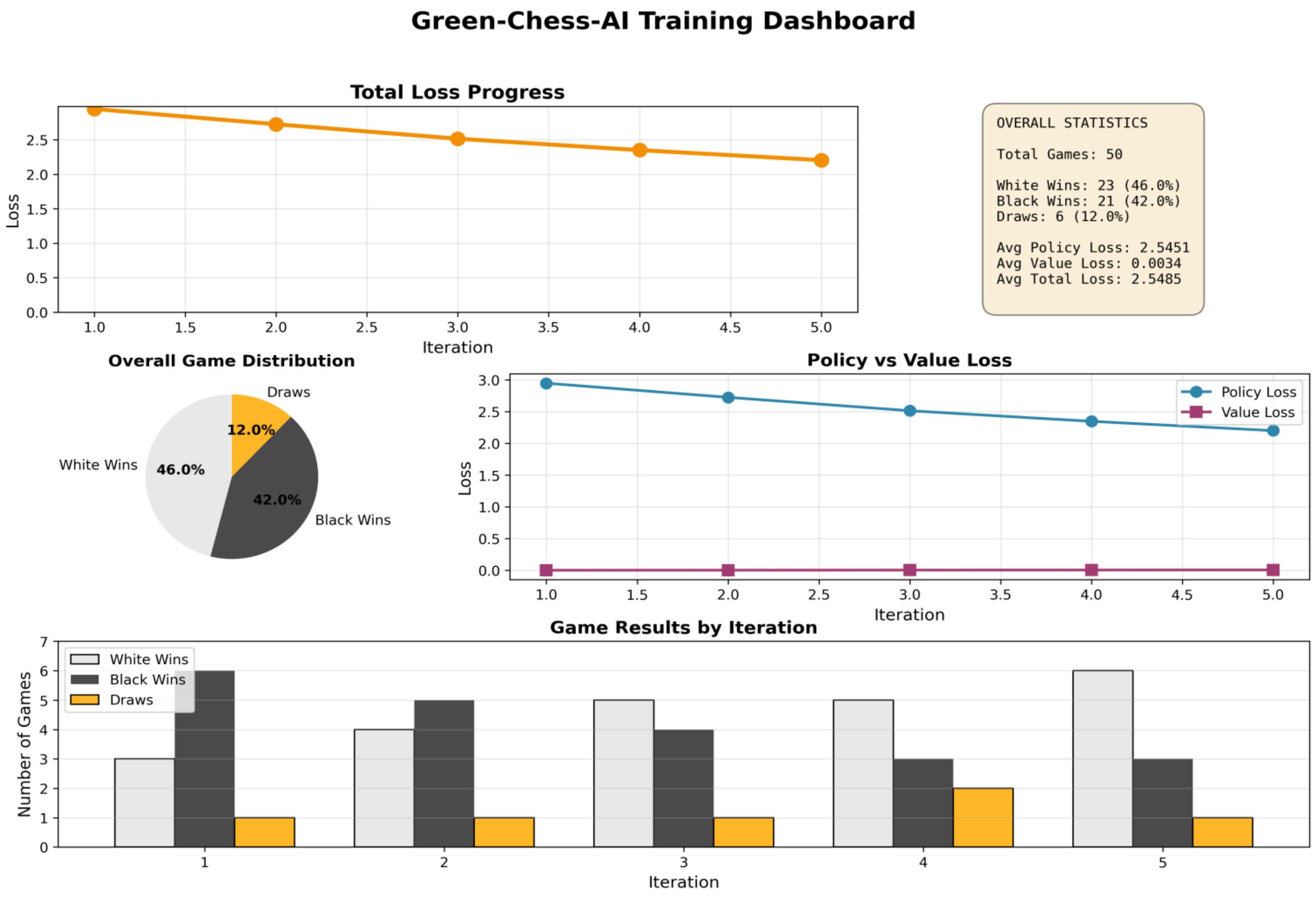
- Why it is Powerful
  - Doesn't need to evaluate every possible move
  - Uses statistics from simulations to guide decisions
  - Perfect for chess: 10^170 possible positions

The Four Phases of Training

- 4 Phases
  - Selection: Traverse the tree using UCT formula
    - Key Formula (UCT):
      - Balances quality + exploration using: $Q(s,a) + c \times P(s,a) \times \sqrt{(N(s))} / (1 + N(s,a))$
      - Higher Q values = exploit good moves
      - Higher uncertainty = explore new moves

  - Expansion: Add new child nodes for unexplored moves
  - Simulation: Play random moves until game ends
  - Backpropagation: Update statistics all the way back to root

Current Analytics



Chess AI Platform: Summary

- **Summary**
  - **Interactive Chess Platform:** Play against friends or an intelligent AI opponent.
  - **Core Features:** Move tracking, game history, and an intuitive user interface.
  - **Full-Stack development:** Seamless integration of frontend, backend, and data management.
  - **Skills Showcased:** Software development Lifecycle, problem-solving, and project development process.
- **Future Directions**
  - **Enhance UX:** More intuitive controls and multiplayer options.
  - **Analytics & Insights:** Track player stats, move patterns, and game analysis board
  - **Scalability:** Store user profiles, game history, and stats
  - **AI Upgrades:** Smarter, more challenging opponents.
- **Statement**
- *"Submitted prior to presenting—demonstrating both results and process."*

ANY Question